Name: _____ Period: ____

Relative Velocity Investigation

Unit 2 · River crossings and crosswinds · Use with the Relative Velocity Lab

Open the Relative Velocity Lab on your device, using the link below or the link on Canvas. Read each part, predict before you run the simulation, then record what you see. You already have every math tool you need from Sections 2.1–2.4: tip-to-tail addition, the Pythagorean theorem, SOH-CAH-TOA, and adding vectors by components. Today you use them on a new idea. Unless a step says otherwise, leave **Hide answers** unchecked.

Simulation: https://shorturl.at/XWHMf (also on Canvas)

$$A_x = A\cos\theta \qquad A_y = A\sin\theta \qquad R = \sqrt{R_x^2 + R_y^2} \qquad \theta = \tan^{-1}\left(\frac{|R_y|}{|R_x|}\right) \qquad v = \frac{d}{t}$$

All students: everyone completes these parts. **Honors · ★ Challenge for general**: honors physics completes these; general physics may try them as a Challenge.

Part 0 · Warm up: tools from Sections 2.1–2.3

All students

1. A car's speedometer reads 25 m/s. A GPS app says the car's velocity is 25 m/s north. What does the GPS tell you that the speedometer does not? Which one is a vector?

2. Add these velocities tip-to-tail: 4.0 m/s north and 3.0 m/s east. Sketch the triangle, then find the magnitude and direction of the resultant.

3. A velocity is 4.0 m/s at 30.0° east of north. Find its north and east components.

Part 1 · Explore: a boat aimed straight across

All students

Open the **River crossing** tab. Set boat speed 4.0 m/s, heading 0° (straight across), current 3.0 m/s, river width 80 m. Uncheck **Show predicted landing spot** for now.

NOTE · WORDS TO KNOW **Heading:** the direction the boat points. **Current:** how fast and which way the water moves relative to the ground. **Downstream** is the way the current flows; **upstream** is against it. **Landing spot:** how far downstream of the start dock the boat reaches the far bank.

4. **Predict.** Before you launch, sketch the path you think the boat will take across the river. Will it land across from the start dock, downstream, or upstream?

5. **Test.** Launch the boat. Describe the path it really takes. Which way does the boat point, and which way does it move? Are those the same?

Keep the boat at 4.0 m/s aimed straight across. Change only the current and read the results panel.

Current (m/s)	Time to cross (s)	Landing spot (m downstream)	Speed relative to the ground (m/s)
0			
1.0			
2.0			
3.0			
4.0			

NOTE · WRITING A CER **Claim:** one sentence that answers the question. **Evidence:** specific numbers from your data. **Reasoning:** the physics idea that explains why the evidence supports the claim.

6. **Claim, evidence, reasoning.** Does a faster current make the boat take longer to cross? Use your data as evidence, and explain why.

Connect to Section 2.2. The boat's velocity and the current are perpendicular vectors, just like the north and east legs of a walk. Check one row of your table with the Pythagorean theorem.

7. Calculate the speed relative to the ground for the 3.0 m/s current. Does it match the simulation?

8. Use $d = vt$ to calculate the landing spot for the 2.0 m/s current. Which speed and which time did you use?

Part 2 · Whose point of view? Reference frames

All students

Launch the boat again, and while it moves, switch **Watch from** to **a raft on the water**. Then switch back to **the shore**.

NOTE · REFERENCE FRAMES A **reference frame** is the point of view a measurement is made from. Example: an airport moving walkway moves at 1.0 m/s, and you walk on it at 1.5 m/s relative to the walkway. A traveler standing beside it sees you move at 2.5 m/s, but a friend riding the walkway next to you sees only 1.5 m/s. Both are correct; they are measured from different frames.

9. From the raft, what path does the boat follow? What happens to the shore and the docks?

10. Each arrow on the boat is one velocity. Match each to its description, using the words *boat*, *water*, and *ground*.

Orange: the boat relative to the _____

Blue: the _____ relative to the _____

Dark: the boat relative to the _____

11. A velocity is always measured relative to something. Fill in the subscripts so the equation matches the tip-to-tail diagram.

$v_{\text{boat, ground}} = v_{\text{boat, }_____} + v_{_____\text{, ground}}$

What pattern do you notice in the subscripts?

Part 3 · Predict, then test

All students

Check **Hide answers** so the simulation keeps its results to itself. Set the river width to 120 m, the boat to 3.0 m/s straight across, and the current to 2.0 m/s. Do the calculations before you launch.

NOTE · DESCRIBING A DIRECTION ON A RIVER Give an angle and a reference, such as 25° downstream of straight across. Sketch with "across" pointing up the page and "downstream" pointing right, just like the simulation.

12. Find the boat's speed relative to the ground and its direction (as an angle downstream of straight across).

13. Find the time to cross and the landing spot.

14. Launch and compare with the simulation. A classmate divided 120 m by 3.61 m/s to get the time. Explain what went wrong.

Part 4 · Planes: the same idea in the air

All students

Open the **Plane in the wind** tab. The airport is 400 km due north. Set airspeed 200 km/h, heading due north, wind 60 km/h blowing toward due east.

NOTE · WORDS TO KNOW **Airspeed:** the plane's speed relative to the air. **Ground speed:** its speed relative to the ground. **Heading:** the direction the nose points. **Headwind:** wind blowing against the plane's motion. **Tailwind:** wind blowing the same way. **Crosswind:** wind blowing sideways to the path. In physics a wind's direction is the way it blows *toward*. Weather reports name where it comes *from*, so a "north wind" blows toward the south.

15. Which velocity plays the role of the boat, and which plays the role of the current? Write the chain equation for the plane using *plane*, *air*, and *ground*.

16. **Predict.** Calculate the ground speed, the direction of travel, and how far from the airport the plane will be when it reaches the airport's latitude. Then take off and check.

Now try wind along the route. Keep the heading due north. A **headwind** blows toward due south; a **tailwind** blows toward due north. Record the time to reach the airport.

Wind	Ground speed (km/h)	Time to the airport
none		
50 km/h headwind		
50 km/h tailwind		
100 km/h headwind		
100 km/h tailwind		

Connect to Section 2.1. When vectors point along the same line, adding them is just adding signed numbers. Treat north as positive.

17. A pilot flies 400 km north into a 100 km/h headwind, then turns around and flies home with that same wind as a tailwind. Does the wind's effect cancel over the round trip? Compare the total time with a round trip on a calm day.

Part 5 · Aim upstream to land straight across

Honors · ★ Challenge for general

Back on the **River crossing** tab: boat 4.0 m/s, current 3.0 m/s, width 80 m, **Hide answers** off. Turn the heading upstream and record the results.

NOTE · MEASURING THE HEADING In the simulation the heading θ is measured from straight across. 0° points straight at the far bank, and a larger upstream angle turns the boat more toward the oncoming current.

Heading (° upstream)	Time to cross (s)	Landing spot (m; + downstream, – upstream)
0		
15		
30		
45		
60		

18. Between which two headings does the boat land directly across? Use the sliders to find the heading to the nearest half degree.

Connect to Section 2.3. Resolve the boat's velocity into an across component and an upstream component, measuring θ from straight across.

NOTE · FINDING AN ANGLE FROM TWO SIDES $\theta = \sin^{-1}\left(\frac{\text{opposite}}{\text{hypotenuse}}\right), \theta = \cos^{-1}\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right), \theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$. Pick the one that uses the two sides you know.

19. Write the boat's across component and upstream component in terms of $v_{\text{boat, water}}$ and θ. To land directly across, what must the upstream component equal? Use that to calculate θ.

20. Calculate the time to cross at that heading. Why does landing straight across always take longer than aiming straight across?

21. **Go further.** Set the current to 4.0 m/s and the boat to 2.0 m/s. Can the boat land straight across? Test headings from 0° to 60° upstream and find the one with the smallest drift. Explain why you can't do better.

Part 6 · A crosswind at an angle

Honors · ★ Challenge for general

On the **Plane in the wind** tab, set airspeed 200 km/h, heading due north, airport due north, and wind 80 km/h blowing toward 30.0° N of E (drag the wind slider until it reads 30.0° N of E). These vectors are not perpendicular. Check **Hide answers**.

Connect to Section 2.4. To add vectors that aren't perpendicular, break each into components, add the columns, then rebuild the resultant.

NOTE · READING A COMPASS ANGLE 30.0° N of E means: face east, then turn 30.0° toward north. Example: 50 km/h at 40.0° N of W has an east component of $-50\cos 40.0^\circ = -38.3$ km/h (west is negative) and a north component of $+50\sin 40.0^\circ = +32.1$ km/h.

Vector	East component (km/h)	North component (km/h)
plane relative to the air: 200 km/h due north		
air relative to the ground: 80 km/h at 30.0° N of E		
plane relative to the ground (sum)		

22. Find the plane's ground speed and direction. Then find the time to reach the airport's latitude and how far east of the airport the plane ends up. Uncheck **Hide answers** and take off to check.

23. **Correct for the wind.** What heading makes the plane's east component of ground velocity zero? Calculate it, set it, and fly. Then compare the flight time with the uncorrected flight.

Wrap up

All students

24. A friend says, "To get somewhere in a current or a wind, just point at it." Use what you saw in the simulation to explain when that works and when it doesn't.

25. **Exit question.** Mr. Peck swims at 1.5 m/s relative to the water, aimed straight across a 45 m river with a 0.80 m/s current. Find the time to cross, the drift, and the speed relative to the shore. *Honors:* what heading lets Mr. Peck land straight across, and how long does that take?